From Behavior Cloning to Online RL

Initial performance, success coverage, reliability, and compute allocation

Graduate research proposal | Two-student team | $500 maximum compute expenditure

Research and pricing checked: 23 October 2026. **Status:** proposed experiments; no model training or throughput measurements have been performed for this document.

Abstract

This project studies how the initial distribution of policy success across problems affects improvement through online learning. Two policies can have the same single-attempt accuracy while differing substantially in the breadth and consistency of their successes. We will construct different initializations of Qwen2.5-0.5B-Instruct through supervised training, compare a precisely specified GRPO configuration with online filtered supervised learning, and measure pass@1, pass@4, and pass^4 under equal online-training and equal total-training costs. The primary environment is an arithmetic construction task requiring two, four, or six decisions and providing one terminal binary reward. A restricted-demonstration intervention tests whether the structures shown during behavior cloning affect later learning. A smaller GSM8K experiment checks whether initialization trends extend to natural language reasoning. The full plan contains 36 training branches across three seeds, with separate allocations for development, supervised initialization, evaluation, and failures. Published evidence motivates the model choice but does not establish zero-shot performance or reliability under our protocol. Accordingly, an integrated measurement phase checks task suitability and throughput before the main matrix is frozen. The intended contribution is a controlled characterization of initialization-dependent learning and its costs, not a new optimizer or a universal rule for choosing when to start RL.

Design decisions after evidence review

- Keep the 0.5B model as the primary candidate, but do not infer its zero-shot accuracy or pass^4 from a reported GSM8K benchmark score.
- Retain the 2/4/6-decision task mixture, with development-only checks for floor, ceiling, format failure, and truncation. Describe it as controlled multi-step reasoning, not evidence about very long robotic episodes.
- Retain eight attempts per test problem. Add a preselected 96-problem, 16-attempt arithmetic audit if the measured evaluation budget supports it; report pass^2 and valid rare-event bounds when pass^4 is poorly resolved.
- Make GRPO's loss normalization explicit. Use a constant completion-length denominator to avoid adding an inverse-length weighting to the horizon comparison [14, 18].
- Treat three seeds as limited evidence about training variability. Preserve paired comparisons and uncertainty; do not promise small-effect significance or a conference-level result in advance.

Contents and protocol map

The proposal moves from the research question to an executable protocol. Sections 1-3 define the outcomes and research gap; Sections 4-7 assess the model and specify data and learning; Sections 8-10 define execution, inference, and cost. The final table is the complete ledger of funded comparisons, diagnostics, and fixed controls.

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How the questions map to experiments

Question	Main evidence	Supporting check
Which initial success profiles improve?	A: BC-level comparison; B: demonstration restriction	Per-horizon profiles and independent retention analysis
Which online objective is effective?	A/B: GRPO versus filtered supervision	C: continued BC; D: group-size allocation
Which complete pipeline is efficient?	Equal-online and equal-total cost endpoints	E: limited GSM8K initialization check

The 36-branch plan is a maximum funded scope. The core claims rely on A, B, and C; D and E are secondary and are removed first if measured costs require it. Suitability checks occur inside the project, before the final protocol is frozen.

1. Research question and scope

The central question is: **How does a base policy's pattern of initial success determine whether online learning broadens solvable-task coverage, makes existing successes more consistent, or loses previously reliable behavior?** We also ask how the answer changes with required decision count and the cost of obtaining the initialization.

Behavior cloning (BC) means supervised learning from demonstrated solutions. In this language-model setting, BC and task-specific supervised fine-tuning (SFT) refer to the same training stage. The released starting model is already instruction-tuned; “no BC” means no additional task-specific weight updates. It does not mean random initialization, a pretraining-only model, or a policy with no prior task knowledge [1, 2].

We manipulate supervised exposure and the demonstrated solution structures. These are experimental controls, not the conceptual definition of policy strength. We measure the resulting initial success profile and follow its evolution during online learning. The design identifies effects of the training interventions, but it does not independently randomize accuracy and reliability.

The primary study uses one model, one arithmetic environment, two online objectives, and three training seeds. There is no robotics implementation, broad algorithm tournament, learned reward model, external teacher API, or pretraining experiment. GSM8K is a limited external check. The scope reflects a $500 total compute constraint, including development and evaluation.

1.1 Three hypotheses and their falsifiers

Hypothesis	Test and expected pattern	Evidence against the hypothesis
H1: Initial success structure matters beyond a single accuracy ranking.	Compare initial and final profiles across BC levels; test broad versus restricted demonstrations at two epochs. Broader demonstrations may yield better final coverage, especially at horizons four and six, without the best initial pass@1.	Rankings remain aligned with initial accuracy and the coverage intervention produces no reproducible gain. If profiles do not meaningfully differ, the intended distinction is not tested.
H2: The online objective interacts with initialization.	Compare GRPO with filtered supervision from identical checkpoints. Relate gains to the availability of successes and mixed-reward groups; test group sizes four and eight at fixed responses per update.	The same method has comparable advantages across initializations, or group statistics fail to explain differences. A correlation alone does not establish a causal gradient mechanism.
H3: Accounting for initialization cost can change the preferred pipeline.	Compare equal-online-cost and equal-total-cost endpoints for the same trajectories. Additional BC may help at equal online cost but consume budget needed for online improvement.	Rankings and practical differences remain unchanged because BC is inexpensive or its benefits dominate its cost. This is an informative result, not a failed experiment.

Pass@1 is the prespecified primary scalar ranking metric. Pass@4 and pass^4 are complementary principal outcomes reported for every condition. No composite score will hide a tradeoff. A method dominates another only if it improves the relevant outcomes within uncertainty at the stated cost. The term “best” must always specify the metric, horizon, and cost convention.

2. What performance and reliability mean

For a fixed problem x and fixed stochastic decoding procedure, let $p_{\pi}(x)$ be the probability that policy pi generates a verified correct solution. Expectations below are over held-out problems, weighted equally within each horizon; the overall arithmetic score averages the three horizons equally.

$$P_1 = \mathbb{E}_x[p_\pi(x)], \quad C_4 = \mathbb{E}_x[1 - (1 - p_\pi(x))^4], \quad R_4 = \mathbb{E}_x[p_\pi(x)^4]$$

P_1 is pass@1, or single-attempt accuracy. C_4 is pass@4, the chance of at least one correct solution in four independent attempts. R_4 is pass^4, the chance that all four attempts are correct [7, 8]. "Coverage" therefore means coverage at a finite retry budget, not the mathematical support of the policy. "Reliability" means repeated-sampling consistency on an unchanged problem, not robustness to adversarial inputs, perturbations, or distribution shift.

Hypothetical initial policy	Pass@1	Pass@4	Pass^4	Mixed groups of four
A: p(x)=0.5 for every problem	50.00%	93.75%	6.25%	87.50%
B: p(x)=1 on half the problems and 0 on the rest	50.00%	50.00%	50.00%	0.00%

These are analytic examples, not measured results. Policy A offers broad opportunities to find successes; Policy B is consistent on a restricted subset. High pass^4 alone can therefore reward specialization. The metrics are dependent summaries of one distribution: $P_1^4 \le R_4 \le P_1$ and $P_1 \le C_4 \le 1-(1-P_1)^4$. One aggregate accuracy value does not determine either repeated-attempt metric.

We will use performance-versus-cost plots and coverage-versus-reliability plots, with panels for horizon. A supplementary three-dimensional plot can place final pass@1, final pass^4, and initial pass@1 on its axes, with final pass@4 represented by color. It is a visualization, not evidence of three independent causal axes.

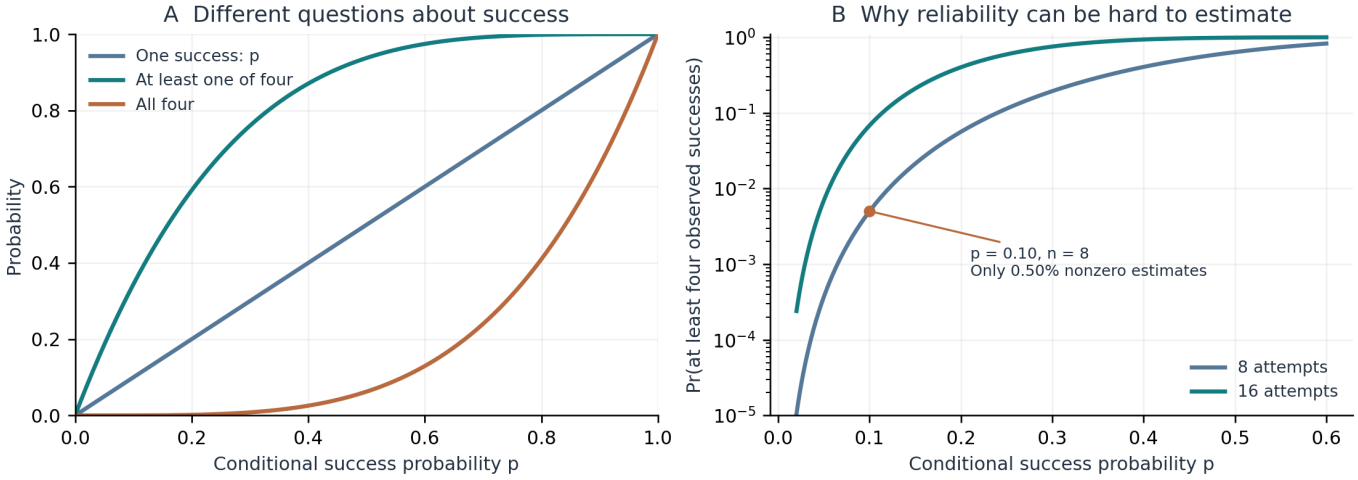


Figure 1. Analytic illustrations, not model results. Left: per-problem success metrics as the conditional success probability changes. Right: the chance that an empirical pass^4 estimate is nonzero, which requires at least four successes among n attempts. Aggregate task metrics additionally depend on heterogeneity across problems.

3. Prior research, remaining questions, and precise contribution

3.1 Initialization and online optimization

Research	What the evidence establishes	What this project adds or does not claim
DeepSeekMath, Shao et al. [3]	Introduces GRPO, replacing a learned value baseline with group-relative rewards, and studies relationships among supervised and policy-optimization objectives.	Provides the algorithmic foundation and filtered-supervision comparison. We test initialization-dependent outcomes under a small fixed budget; neither objective is new.
Quagmires in SFT-RL Post-Training, Kang et al. [4]	Reports that high SFT accuracy can misrank eventual RL outcomes; held-out loss and large-K success can be more useful proxies in its experiments.	Merely reproducing a checkpoint rank reversal is insufficient novelty. We add a structured demonstration intervention, controlled decision counts, repeated-trial reliability, and explicit pipeline cost accounting. Pass@4 is not a substitute for its large-K capacity analysis.
Getting Your LLMs Ready for RL with Lightweight SFT, Li et al. [5]	The ICLR 2026 paper reports distributional forgetting and diversity-based SFT stopping criteria, and proposes Adaptive Early-Stop Loss.	We do not propose a new entropy stopping rule or claim that diversity-aware initialization is unexplored. We characterize coverage/reliability outcomes under fixed budgets without implementing AESL.
SFT Overtraining Predicts Rank Inversion, Aphale and Liu [6]	The DL4C 2026 workshop study links SFT depth, entropy collapse, and GRPO outcome differences in two code-model settings.	This is direct prior art for mixed-group and collapse diagnostics. Our small arithmetic study tests a different intervention and task structure; it does not present the zero-advantage observation as a discovery.
Understanding R1-Zero-Like Training, Liu et al. [18]	Analyzes optimization biases in GRPO-style training and motivates altered loss normalization.	We explicitly choose constant-length normalization and log response length so that an avoidable implementation choice does not obscure the horizon comparison. We retain group reward standardization, so our recipe is not the full Dr. GRPO recipe.

The proposed contribution is a **joint controlled empirical design**: demonstration structures are restricted at matched exposure; task decision counts are explicit; both online objectives are evaluated; and changes in performance, finite-retry coverage, and consistency are charged to a complete training pipeline. We make no exhaustive “first work” claim.

3.2 Evaluation and interpretation of reasoning gains

Research	What the evidence establishes	Consequence for this proposal
Evaluating Large Language Models Trained on Code, Chen et al. [7]	Gives a combinatorial unbiased estimator of pass@K from n independent samples.	Use the estimator with n=8 and K=4, rather than substituting the observed success rate into a nonlinear formula.
tau-bench, Yao et al. [8]	Introduces $pass^K$ as the chance that all K independent trials succeed, averaged across tasks.	Report $pass^4$ alongside pass@1 and pass@4; do not relabel it as general safety or robustness.
Does RL Really Incentivize Reasoning Capacity Beyond the Base Model?, Yue et al. [9]	In the studied settings, small-K improvements can coexist with weaker large-K success, motivating a sampling-efficiency versus coverage distinction.	Track multiple success summaries, but do not infer new capabilities or their absence from eight samples per problem.
RLVR Implicitly Incentivizes Correct Reasoning, Wen et al. [10]	Argues that correct final answers can mask incorrect reasoning and reports different conclusions using a reasoning-validity criterion.	The arithmetic reward verifies the complete operation trace. GSM8K final-answer reward supports an answer-correctness claim only. The wider capability debate remains unresolved by this project.
Deep RL at the Edge of the Statistical Precipice, Agarwal et al. [11]	Demonstrates how few-run evaluation and point estimates can give misleading conclusions.	Show paired seed outcomes and task-level uncertainty separately. Three seeds do not support strong claims about small differences or universal algorithm rankings.

GSM8K originates in work on training verifiers for mathematical answers [13]; the official dataset includes worked solutions [19]. LoRA supplies a parameter-efficient adaptation method [12]. These are components of our implementation, not proposed research contributions. We use primary papers and official documentation; numerical results from unrelated prompts, fine-tuned variants, or community model cards are not substituted for measurements of our exact protocol.

4. Suitability of the model, tasks, and metrics

4.1 What is known about Qwen2.5-0.5B-Instruct

The official Qwen report gives a GSM8K accuracy of 49.6% for the 0.5B instruction-tuned model and 73.2% for the 1.5B version (Table 10) [1]. The 0.5B model card specifies approximately 0.49 billion parameters [2]. Official verl documentation demonstrates function-reward GSM8K training with this model on a one-GPU setup and lists a GPU with at least 24 GB memory as a prerequisite [15]. These facts make it a reasonable candidate, not a validated measurement instrument for our custom task.

Zero-shot evidence limit. The 49.6 score is a publisher benchmark result; it is not an established temperature-1, zero-example, 512-token success rate with our answer parser. The report's explicit four-shot GSM8K description appears in the base-model evaluation section; we do not silently assign that protocol to the instruction-model table. We found no verified result matching the complete proposed protocol. Zero-shot performance on the new trace-construction task is likewise unmeasured. Both must be measured in Phase 0.

The model's size is not itself an obstacle to defining pass@1, pass@4, or pass^4. The obstacle is whether it generates enough correct and inconsistent solutions for the metrics to be informative. A larger model could shift the distribution away from a floor, but would consume more resources and might create a ceiling. Moving to 1.5B is not a free remedy and is not a funded scale comparison in this plan.

4.2 Floor, ceiling, and finite-sample limitations

For a problem with homogeneous success probability p, reliability is p^4. At p=0.1 this is 0.0001, or 0.01%, even though pass@4 is 34.39%. With eight attempts, the probability of observing four or more successes is only about 0.50%. An unbiased estimator can therefore be extremely noisy relative to the quantity it estimates.

Assumed per-problem p	Pass@4	Pass^4	Pr(c≥4), n=8	Pr(c≥4), n=16
0.05	18.55%	0.000625%	0.037%	0.70%
0.10	34.39%	0.0100%	0.50%	6.84%
0.20	59.04%	0.1600%	5.63%	40.19%
0.30	75.99%	0.8100%	19.41%	75.41%
0.50	93.75%	6.2500%	63.67%	98.94%

Table: exact binomial calculations for illustrative homogeneous problems. These are not forecasts from the reported Qwen accuracy; real tasks have different p values.

At high p, pass@4 approaches one quickly and may lose discriminatory power before pass^4 does. Conversely, a reliability floor can remain even when accuracy differences are clear. We therefore retain all three metrics, report pass^2 as a prespecified low-cost diagnostic, and do not treat a zero reliability estimate as proof of zero reliability. No metric is replaced after observing final test outcomes.

4.3 Integrated suitability checks and justified adjustments

Phase 0 is part of the project, not a separate prerequisite pilot. On development examples, first evaluate the untouched instruction model with task instructions but no worked example (true zero-shot with respect to prompt demonstrations). Then evaluate the same model with the one fixed arithmetic format example used in the main study. The latter is a one-example prompted no-BC condition and must not be called zero-shot. For GSM8K, the main prompt contains instructions but no worked solution.

Measure format-validity rate, full success, completion length, truncation, mixed-group frequency, and billed throughput. Phase 0 may train one disposable light-BC copy on 512 development-only demonstrations, charged to the initialization allocation; reset to the original model for the main paths after settings are frozen. Provisional development flags are: over 5% truncation, over 20% format failures after the common format example, or all assessed initializations below 2% or above 95% success on a horizon. These are engineering thresholds chosen for this project, not published universal boundaries. Fix verifier defects first; adjust token caps only if valid demonstrations or outputs are being cut off. Do not grant format rewards or partial-credit rewards to conceal a floor.

Retain horizons 2/4/6 when development results support variation. If horizon six remains at a floor across initializations, preserve it as a reported failure regime but restrict claims about reliability differences to strata with adequate signal. If the entire mixture is uninformative, permit one documented generator adjustment using development data, such as smaller input magnitudes, before regenerating all frozen splits. Do not search for task settings that produce a desired method ranking. If meaningful profiles still cannot be obtained within development resources, report the feasibility limit and narrow the claim; do not silently switch models or spend beyond $500.

Keep n=8 for all primary test estimates. Subject to the measured $90 evaluation envelope, preselect 32 test problems per horizon independently of outcomes and collect eight additional attempts on that same 96-problem panel for every main initial and equal-total-cost final policy. This n=16 audit checks sensitivity to sampling noise without changing the primary panel or claiming to resolve extremely rare reliability events. Its inclusion is decided before final test generation.

5. Task construction, data, and demonstration intervention

5.1 Arithmetic environment and reward

The prompt supplies n integers and a target. A completion is an ordered list of operations of the form “a + b = c”, “a - b = c”, or “a * b = c”. A deterministic verifier maintains a multiset of available values, consumes operands, checks exact integer arithmetic, and inserts the result. Repeated numeric values are allowed only to the extent present in the multiset. A successful trace has exactly n-1 valid operations and leaves the target as its only value. Whitespace may vary; extra claims, missing operations, invalid arithmetic, illegal reuse, and truncation fail. Use a small parser, not execution of arbitrary generated code.

At the policy level, the state is the prompt plus the generated prefix and the action is the next token. The model emits the entire trace before verification; it receives no intermediate environment correction. The controlled arithmetic decision count is therefore distinct from the token-level RL horizon, which varies and is logged. This is autoregressive sequence-generation RL with an exact terminal verifier, not an interactive tool-use or robotic environment.

For example, with inputs 2, 3, 4 and target 20, the two-step trace “2 + 3 = 5; 5 * 4 = 20” succeeds. Reward is one only after the complete trace is verified; all other completions receive zero. Log error categories and valid-prefix length for analysis, but never feed those diagnostics into the training reward. Multiple valid traces may solve the same problem, so online learning can find solutions absent from the demonstrations.

Generate tasks constructively by sampling a full binary expression tree, assigning input integers and operators, computing its target, and retaining a valid trace. Initial generator values are inputs 2-20, operators +, -, *, nonzero positive targets of magnitude at most 1,000, and intermediate absolute values at most 10,000. Publish the exact acceptance rules, operator frequencies, target bins, and rejection counts. Reject zero-producing subexpressions and trivial multiplicative identities in the demonstrated construction. This filter does not prove that no alternative shortcut exists. Apply acceptance constraints uniformly to train, development, and test data; the verifier accepts every trace satisfying the published task rules.

The prompt uses a fixed template and a common worked format example from a separate development pool. Randomize input presentation order with a recorded seed. Deduplicate using

(sorted input multiset, target), not surface prompt text, to prevent permutation leakage. Training solutions and targets are never inserted into held-out prompts.

5.2 Frozen splits

Dataset	Arithmetic task	GSM8K check	Role
Development	Separate generator seed; 144 audit problems and 512 disposable BC demonstrations	128 held-out training examples, also used for validation	Engineering calibration; never final evidence
BC demonstrations	2,048 broad examples; separate matched restricted set	2,000 training examples with worked solutions	Supervised updates only
Online pool	4,096 broad problems, disjoint from both BC sets	4,096 remaining training problems	Prompts and verifier feedback; reference solutions hidden
Validation	144 separate problems: 48 per horizon	128 as above, disjoint from BC/online	Fixed monitoring and permitted selection
Test	288 problems: 96 per horizon	256 official test problems	Locked final evaluation

Arithmetic BC and online splits are as balanced across horizons as their sizes permit. The same data manifests are shared across training seeds; seeds vary optimization and sampling, not the test distribution. Consequently, three seeds do not estimate variability across independent training-dataset constructions. The main study uses a single fixed broad distribution, including both sequential and branching expression constructions.

For GSM8K, use the official worked solutions, remove calculator markup consistently, and create disjoint training-index partitions [13, 19]. Verify actual split sizes and hashes before training. Normalize exact numerical answers using a frozen rule accepting a final "#### number" or a final boxed number; conflicting answers are invalid. Report strict marker compliance separately. A correct final number is not evidence that every reasoning step is correct. Existing pretraining exposure to GSM8K cannot be ruled out; this is a transfer-of-trend check, not a contamination-free reasoning benchmark.

5.3 Broad versus restricted demonstrations

The broad construction includes sequential chains and branching subexpressions; the restricted construction demonstrates only chains that repeatedly combine a running result with a new input. For five inputs, a chain may have form ((((a+b)*c)-d)+e), while a branching construction may have form ((a+b)*(c-d))+e. Both demonstrations are correct, and both policies face the same broad online and test distributions.

Match example count, horizon proportions, input and target bins, and operator histograms. Target a difference of at most 5% in total supervised completion tokens; if that cannot be achieved, report the residual and show cost/token-normalized comparisons rather than claim exact matching. Use two epochs in both conditions. The restriction is meaningful primarily at four and six decisions; two-operation binary trees provide little structural contrast.

This intervention changes the training-data distribution and may change difficulty as well as demonstrated structure. A restricted generating expression does not prove that its problem has no

branching solution. We therefore claim an effect of this specified data intervention, not a pure causal effect of “coverage” isolated from every latent property. If initial pass@1 differs by at most five percentage points on validation, identify that pair as approximately accuracy-matched, while retaining the actual difference and uncertainty. This descriptive label is not randomization of reliability.

6. Initial policies and fixed implementation

For each of three seeds (11, 22, 33), train one broad BC path and save I_0 (no updates), I_{light} (0.25 epochs), and $I_{extended}$ (2 epochs). Train the restricted two-epoch path separately. Stratify the first quarter-epoch by horizon so its shorter exposure is not accidentally dominated by easy problems. Quarter-epoch versus two-epoch training changes both distinct-example exposure and update count; the study does not disentangle those two components.

All models use the same pretrained revision and tokenizer. Adopt LoRA rank 16, alpha 32, dropout zero, targeting attention q/k/v/o and MLP gate/up/down projections, with all original model weights frozen [12]. Use BF16, completion-only supervised loss, gradient accumulation, and a fixed chat template. Verify trainable parameter names and counts. Reset online optimizer state when branching from a BC checkpoint; copy adapter weights exactly. I_0 has identical predictions across seeds before online training, so do not pretend its three copies are independent initial models.

Configuration	Prespecified implementation
Supervised objective	Standard cross-entropy over completion tokens; prompt tokens masked
Supervised optimizer	AdamW; initial learning rate 1e-4; weight decay 0; gradient norm cap 1
Online calibration	At most 5e-6 and 1e-5 learning rates per method on the same development initialization; freeze one per method afterward
Online rollout batch	128 completions: 32 prompts x 4 samples in the main matrix
Online optimizer	AdamW, weight decay 0, gradient norm cap 1; one accumulated optimizer step per fresh batch
Decoding	Temperature 1.0; top-p 1.0; top-k filtering disabled; independent draws; repetition penalty 1
Output caps	256 arithmetic tokens; 512 GSM8K tokens; EOS stopping; truncation is a scored failure
GRPO choices	Group-mean baseline; population group standard deviation plus 1e-6; KL coefficient 0; constant-length loss denominator; no dynamic group filtering
Data reuse	No replay and no repeated optimization epochs on an online batch
Environment	Pin Transformers, PyTorch, PEFT, TRL, CUDA, model revision, and full generation configuration after integration

The learning rates are proposed starting choices, not experimentally established optima. Calibration gets the same dollar allowance and selection rule for both methods: reject numerical failures and choose higher development pass@1, with the smaller rate breaking ties. No

initialization-specific tuning is allowed. Record configurations that fail instead of silently replacing them. Microbatch sizes may change to fit memory; the effective batch and loss reduction must remain fixed.

7. Online objectives and the mechanism being tested

7.1 GRPO configuration

For each prompt, sample G completions from the current policy and verify binary rewards. Let the group advantage be:

$$A_i = \frac{r_i - \bar{r}}{\sqrt{G^{-1}\sum_{j=1}^{G}(r_j - \bar{r})^2 + 10^{-6}}}$$

For a mixed group, successes receive positive advantage and failures negative advantage. All-failure and all-success groups have zero relative task-reward advantages [3, 14]. Use the usual token probability ratio to the rollout policy and the clipped surrogate with clipping radius 0.2, summed over completion tokens and divided by the fixed maximum completion length L_{max} and batch completion count. Keep the KL coefficient zero; there is no separate reference-model regularizer or critic. Label this as a reference-free, group-standardized GRPO configuration with constant-length normalization, not the entire family of GRPO algorithms.

$$\mathcal{L}_G = -\frac{1}{QL_{max}}\sum_{i=1}^{Q}\sum_{t}\min\{\rho_{i,t}A_i,\ \mathrm{clip}(\rho_{i,t},\ 0.8,\ 1.2)A_i\}$$

Here Q=128 is the rollout batch size, each A_i is computed within its prompt's group, and rho is the current-token probability divided by the rollout-policy probability for the same prefix. Masks exclude prompt and padding tokens; all generated completion tokens, including invalid or truncated outputs, retain their terminal reward.

Normalization adjustment. Current TRL documentation exposes multiple loss definitions and explicitly warns about response-length bias in its original sequence-normalized GRPO loss [14]. Because horizon changes output length, use the constant denominator associated with its dr_grpo loss setting, while retaining group standardization. This differs from the earlier conceptual sequence-normalized example and from the full unstandardized Dr. GRPO recipe [18]. Validate the loss against a hand-computed batch; do not rely on a changing library default.

With exactly one full-batch optimizer step per rollout batch, policy ratios equal one when that step's gradient is evaluated, absent numerical mismatch. Clipping is then inactive. The main comparison concerns group-relative credit versus positive-only imitation, not the benefits of repeated PPO-style clipped updates. Log ratio deviations to detect stale rollout weights or an implementation that accidentally performs extra steps.

7.2 Online filtered supervised learning

Draw the same number of fresh completions from the same prompt distribution, retain verified correct outputs, and take one optimizer step to increase their likelihood. Use no replay. For successful subset S, define:

$$\mathcal{L}_{\text{filtered}} = -\frac{1}{|S|L_{\max}} \sum_{(x,y)\in S} \sum_{t} \log \pi_\theta(y_t \mid x, y_{<t})$$

The constant token denominator matches the online GRPO length convention. Averaging over successes is intentional; it defines the baseline and can yield different update scales as the success count changes. The common learning-rate calibration does not make the objectives mathematically identical. Skip the optimizer step when S is empty. For GRPO, skip a batch with no nonzero advantages so stale optimizer momentum does not create an unexplained update.

Filtered supervision can learn from all-success groups, while group-relative task reward cannot distinguish their members. It does not explicitly suppress failed sampled solutions. The methods' generated datasets naturally diverge as their policies change. Pair initial weights and prompt streams where feasible, but do not force identical completions or call a fixed offline sample comparison “online” learning.

7.3 Mixed rewards explain an opportunity, not a guarantee

For independent trials on a fixed prompt, the probability of a mixed-reward group is:

$$M_G = \mathbb{E}_x[1 - p_\pi(x)^G - (1 - p_\pi(x))^G], \qquad M_4 = C_4 - R_4$$

This algebraic identity is not a research contribution. It connects success profiles to the availability of nonzero relative advantages. At fixed Q generated responses, the expected number of mixed groups is (Q/G) M_G, not merely M_G. Larger groups increase the mixed fraction per prompt but reduce distinct prompt exposure. Mixed groups can still produce poor or interfering gradients; shared parameters can improve a prompt that never directly yielded a success. We will report descriptive associations and a targeted sampling intervention, not claim that a mixed-group count uniquely determines learning.

Log successful outputs, mixed/all-success/all-failure groups, entropy averaged over generated tokens, completion lengths, error categories, gradient norms, optimizer steps, generated and processed tokens, and dollar cost. On validation demonstrations, also log held-out supervised loss as a prior-work comparator [4]. These diagnostics are not extra reward terms or a trained checkpoint selector.

8. Execution order and full experiment sequence

The experiment is run in the order below. All final test outputs remain hidden until model configurations and inclusion rules are frozen. Development checks are part of Week 1, not a request to conduct an additional project before this proposal can proceed.

Step	Phase and timebox	Concrete actions and exit condition
0	Integration and suitability, Week 1	Build generator and verifier; audit zero-shot versus common-format prompting; test arithmetic edge cases; profile L4 and L40S if available; verify parameter updates, loss reduction, and restart. Freeze generator, splits, parser, hardware, configs, and evaluation envelope.
1	Initialization, Week 2	Produce three broad BC paths and three restricted paths; record initial profiles and held-out loss. Fork exact checkpoints and fresh online optimizer states. Establish continued-supervision paths on the same demonstrations.
2	Experiment A, Week 3	Run 3 initializations x 2 objectives x 3 seeds = 18 branches. Interleave paired methods in execution order. Save cost-indexed checkpoints and measure validation at fixed checkpoints.
3	Experiments B and C, Week 4	Run 6 restricted-initialization online branches and 3 continued-BC baselines. Compare with already-computed broad controls. Test coverage intervention and the value of online data collection.
4	Experiment D, Week 4	From broad light BC, run 3 G=8 GRPO branches; reuse G=4 controls. Keep 128 completions per update, so distinct prompts change from 32 to 16.
5	Experiment E, Week 4	Run GSM8K light and extended BC followed by GRPO, 3 seeds each. Use a $6 online cap per branch and the fixed natural-language verifier.
6	Locked evaluation and analysis, Week 5	Evaluate initial and specified cost endpoints; compute three primary metrics, uncertainty, retention/discovery diagnostics, and cost curves. Release code, manifests, configurations, raw evaluation outcomes, cost ledger, and report.

8.1 Experiment A: initialization by online objective

The full factorial is $\{I_0, I_{light}, I_{extended}\}$ x {GRPO, filtered supervision} x {11,22,33}. Each branch has at most $8 of online training. Compare paired metric changes from initialization, final metric rankings, and method differences separately by horizon. The method difference at initialization i is delta_i = score(GRPO,i) - score(filtered,i); differences between delta values describe an initialization-by-method interaction. With three seeds and few initializations, these are effect estimates and plots, not a reliable high-dimensional predictor fitted to dozens of correlated metrics.

A reversal between initial and final pass@1 rankings supports the narrow statement that initial accuracy misranked those checkpoints. If the initial profiles overlap too strongly or the effects are small relative to uncertainty, the experiment is inconclusive about profile-dependent learning. It cannot establish that reliability itself caused an outcome independently of the BC intervention.

8.2 Experiment B: restricted demonstration structures

Train online from the restricted two-epoch initialization with both methods and three seeds (six branches, $8 each). Compare against broad $I_{extended}$ runs from A. The primary contrast is final broad-test pass@4 at equal total cost, with pass@1 and $pass^4$ alongside it. Report horizons four and six separately and horizon two as a limited-contrast stratum. The design tests coverage restriction at extended BC only; it is not a full coverage-by-BC-duration factorial.

8.3 Experiment C: continued supervised training

Continue training on the original broad 2,048 demonstrations for each seed, without new online samples, up to $8 of standalone pipeline cost. Reuse the initial BC path rather than pay twice for identical work. Save cost-indexed checkpoints. Report the fixed-budget final checkpoint and a validation-selected checkpoint using the lowest held-out supervised loss, with ties favoring the earlier checkpoint. Charge the latter for all training and validation required to select it, even if the returned weights were saved earlier. This prevents an intentionally overtrained baseline and prevents refunding search cost.

If continued BC matches online methods, the project has identified a regime where online sampling and reward-based optimization provide little benefit under the measured budget. This does not show that RL is unnecessary when demonstrations are unavailable or at other scales.

8.4 Experiment D: sampling group size

Use broad I_{light} and GRPO with G=8 for three seeds, at $8 each. Compare with the G=4 light-BC controls from A. Both use 128 generated responses per update: 16 prompts x 8 versus 32 prompts x 4. Report results against dollars and generated tokens. Prompt diversity, reward normalization, and effective update structure change together; this is a sampling-allocation ablation, not proof of a single isolated gradient mechanism. More mixed groups per prompt without better final scores would weaken the practical value of that diagnostic.

8.5 Experiment E: natural-language check

On GSM8K, create 0.25- and 2-epoch initializations and run the frozen GRPO recipe for three seeds each, with a $6 online cap per branch. Measure initial and final pass@1, pass@4, and $pass^4$ using n=8 on the 256-problem test subset. Evaluate equal-online-$6 and equal-total-$6 checkpoints where feasible. Do not repeat the entire demonstration and algorithm matrix. Agreement supports transfer of an initialization trend; disagreement identifies a domain limitation. Neither outcome validates cross-modal generality or a causal GSM8K horizon effect.

8.6 Run count and priorities

Block	Branches	Existing controls reused	Cap
A: main comparison	18	Initial BC checkpoints	$144
B: restricted demonstrations	6	Broad extended conditions in A	$48
C: continued BC	3	Broad BC path	$24
D: G=8	3	Broad light G=4 conditions in A	$24
E: GSM8K	6	Two initializations per seed	$36
Total downstream branches	36	BC creation budget is separate	$276

These training branches measure learning within a budget; convergence is not an inclusion criterion. If infrastructure or evaluation costs exceed the development forecast, protect A, B, C, the three paired seeds, and primary evaluation first. Omit D, then E, before reducing the core

comparison. Record any scope reduction before inspecting final test results; retain the omitted conditions in the final experiment manifest as not run.

9. Evaluation, uncertainty, and failure interpretation

9.1 Repeated-sampling estimators

For N test problems, n=8 independent completions each, and c_x correct completions, compute:

$$\widehat{P}_1 = \frac{1}{N}\sum_x \frac{c_x}{n}$$

$$\widehat{C}_4 = \frac{1}{N}\sum_x \left[1 - \frac{\binom{n-c_x}{4}}{\binom{n}{4}}\right], \quad \widehat{R}_4 = \frac{1}{N}\sum_x \frac{\binom{c_x}{4}}{\binom{n}{4}}$$

Set choose(a,4)=0 when a<4. The combinatorial ratios count four-element subsets with at least one success or all successes, avoiding the plug-in bias of nonlinear functions of c_x/n [7, 8]. Use identical prompts, caps, stochastic decoding settings, and scoring across conditions. Greedy decoding is an auxiliary pass@1 measurement only: repeating an identical deterministic output does not estimate stochastic repeated-trial reliability.

Use independent random draws within a problem and retain task, policy, seed, attempt index, text, token count, parser result, reward, and finish reason. Do not serve repeated attempts from an output cache. Initial development audits use eight attempts per problem; intermediate validation monitoring uses one attempt per problem and is not a reliability estimate. The n=16 audit uses the original eight plus eight new independent attempts on the fixed panel; report n=8 and n=16 estimates on the same panel. Its overlapping samples make the estimates dependent, which must be respected in a paired sensitivity analysis.

9.2 Separate two kinds of uncertainty

Compute each metric separately for each training seed and show all seed points. For uncertainty over task sampling, use a paired, horizon-stratified bootstrap with 10,000 resamples of whole problems, keeping each problem's attempts together and using the same resampled task indices across policies. Report these intervals as conditional on the trained checkpoints. A hierarchical seed-and-task bootstrap may be included, but with only three seeds it is unstable and must not be presented as decisive evidence.

Three seeds cannot establish robust small-effect superiority. For example, a two-sided sign-only test on three paired differences has minimum attainable p-value $2/2^3 = 0.25$. More test completions do not create additional independent training runs. This project emphasizes effect sizes, consistency of direction, and practical differences, following the few-run evaluation concern in [11]. Avoid a large collection of uncorrected significance tests.

Because each per-problem metric estimator lies in [0,1], its standard deviation across independently sampled problems is at most 0.5. A rough normal-approximation 95% half-width bound is therefore 1.96/(2 sqrt(N)): about 5.8 percentage points for N=288 and 10.0 points for N=96. These are conservative planning scales, not guaranteed confidence intervals or a power calculation for the paired comparison. Actual precision depends on task heterogeneity, paired covariance, and seed variability.

9.3 Handle rare reliability without false precision

If every observed pass[4] contribution is zero, a naive empirical bootstrap can return a misleading zero-width interval. As a conservative check, preselect one quartet per test problem and treat its all-success indicator as one Bernoulli trial per independently sampled problem. With zero such successes, the one-sided 95% upper bound is 1-0.05^(1/N), approximately 1.03% for 288 problems and 3.07% for 96. For horizon-stratified analysis, apply bounds within each stratum and use simultaneous bounds if aggregating them. This check sacrifices information but avoids pretending that repeated failures prove zero reliability.

Eight failures on an individual problem also do not establish impossibility: if p=0.1, zero successes occur with probability 0.9^8 = 43.05%. The exact one-sided 95% upper bound from 0/8 is about 31.23% for that problem. A high pass@4 or a newly observed success likewise does not establish a previously absent capability. Claims are limited to finite-sampling success under the tested protocol.

9.4 Discovery, consolidation, and preservation

For a secondary problem-level analysis, use four initial attempts only to assign “no observed success,” “intermittent,” or “all observed successes” groups. Use the other four attempts to estimate independent initial accuracy within those groups, and compare with fresh final attempts. This sample split reduces regression-to-the-mean artifacts from grouping and evaluating with the same noisy initial observations. Aggregate metrics still use all eight samples; this diagnostic does not replace them.

Improvement in the first group suggests increased accessibility of success, improvement in the middle suggests consolidation, and deterioration in the last suggests loss of previously consistent behavior. The labels refer to observations, not true p=0 or p=1. Track format validity and operation validity alongside correctness. In GSM8K, report answer correctness without asserting that the reasoning path was verified.

10. Cost accounting and feasibility

10.1 Two cost conventions

Equal-online-cost comparisons hold the spending after initialization fixed. Equal-total-cost comparisons include initialization and any checkpoint selection:

$$C_{\text{pipeline}} = C_{\text{BC}} + C_{\text{online}} + C_{\text{selection}}$$

Use $8 as the main arithmetic comparison point ($6 for GSM8K). Save checkpoints at the corresponding online allowance B-C_BC-C_selection as well as at the equal-online endpoint. If the allowance is negative, that pipeline is infeasible at that total budget; do not extrapolate backward or omit its BC charge. Lower-cost validation checkpoints describe learning curves, but the full repeated test evaluations are concentrated at the prespecified endpoints.

The project ledger counts actual shared computation once. The standalone ledger charges each pipeline the BC required to produce its initialization. The cost of producing the shared pretrained, instruction-tuned checkpoint is outside this post-training budget; loading, online generation, updates, verification, necessary selection, and billed pipeline overhead are included. Research-only

evaluation is charged to the $500 project ledger but not to a fixed-schedule training pipeline. Selection or early stopping that uses evaluation must pay for it in the standalone ledger too.

Report effective dollars, GPU type, GPU seconds, CPU/RAM usage, generated tokens, total processed tokens, and optimizer updates. Dollars are the primary constraint; token counts help distinguish implementation throughput from sample efficiency. Match checkpoint budgets to within one optimizer batch and report actual expenditure. Keep the hardware and inference backend fixed within a comparison after development. Partition runs across the two students' credit balances if necessary; do not assume credits can be transferred or that additional free credits are available.

10.2 Budget envelope

Expense	Maximum
Integration, suitability checks, throughput and limited configuration calibration	$35
Supervised initialization paths, including GSM8K	$35
A: 18 main online branches x $8	$144
B: 6 restricted-demonstration branches x $8	$48
C: 3 continued-supervision branches x $8	$24
D: 3 group-size branches x $8	$24
E: 6 GSM8K branches x $6	$36
Evaluation and analysis compute	$90
Reserve for retries, overruns, storage, and mandatory evaluation	$64
Total	$500

Modal lists L4 at $0.000222/s ($0.7992/h) and L40S at $0.000542/s ($1.9512/h) [16]. At two physical CPU cores and 16 GiB RAM, their listed resource rates add $0.222192/h, giving illustrative totals of $1.021392/h and $2.173392/h. CPU and RAM are billed at the higher of requested or actual usage [17]; storage, larger memory use, additional containers, and other charges are not included in those illustrative rates. Use standard Functions for the estimate; do not assume Sandbox pricing is identical.

At the illustrative L40S total, an $8 run allows about 3.68 billed hours; on the L4 it allows about 7.83 hours. This does not tell us how many useful updates fit. LoRA lowers trainable-state requirements, but autoregressive generation, activations, and the inference engine still consume resources [12]. Choose hardware by measured end-to-end cost per useful batch, not hourly price alone. No throughput or final accuracy estimate in this document is an observed result.

10.3 Evaluation is a substantial workload

The upper-count arithmetic evaluation manifest contains 10 distinct initial checkpoints (one shared I_0 and three each of light, extended, restricted), 54 endpoints from 27 online runs with two cost conventions, and three continued-BC endpoints: 67 policy evaluations. At 288 problems x 8 attempts, this is 154,368 completions. The optional n=16 audit adds 34 policy evaluations x 96 problems x 8 new attempts = 26,112 completions, for 180,480 arithmetic completions. Identical initial or cost endpoints may be reused rather than regenerated.

The GSM8K manifest has six initial policies and twelve endpoints, giving 18 x 256 x 8 = 36,864 completions. These counts exclude Phase 0, intermediate validation, and the optional selected supervised-baseline endpoint; all are included in the measured evaluation forecast before final scope is locked.

For an illustrative arithmetic mean completion length of 64 tokens and GSM8K mean of 256, these full endpoint manifests generate about 21.0 million tokens. At their hard caps, they can generate up to about 65.1 million. These are arithmetic scenarios, not measured length distributions. At the illustrative L40S rate, $90 buys about 41.4 hours, requiring at least about 141 generated tokens/s for the mean-length scenario or 437 tokens/s for the cap scenario, before allowing for additional evaluations. “Effective throughput” must include prefill, model loading, verifier time, and idle billed time.

Profile representative complete batches and extrapolate from actual lengths and billed time. Add a 20% scheduling margin to the measured forecast, with the margin funded by the evaluation allocation and reserve. If it does not fit, reduce optional audit evaluations and secondary blocks D/E before primary n=8 evaluation or paired seeds. If even the core plan does not fit, lower all paired core online caps consistently before main training and report the revised budget estimand. Never keep the favorable member of an unfinished pair.

11. Expected findings, claim boundaries, and deliverables

Possible result	Supported interpretation	Interpretation to avoid
Initial and final accuracy rankings differ	Initial accuracy misranked the tested checkpoints	A universal RL-readiness predictor has been discovered
Broad demonstrations improve final coverage at matched budgets	This demonstration intervention helps downstream coverage	Reliability was independently randomized or every latent difficulty was controlled
Pass^4 increases while pass@4 decreases	Success became more concentrated and consistent on part of the distribution	The policy became universally better or safer
Filtered supervision matches or exceeds GRPO	Positive-sample imitation is competitive under this recipe and budget	All RL algorithms are unnecessary
Mixed-group frequency rises without improved learning per dollar	The diagnostic is not sufficient to justify a sampling allocation	Nonzero relative advantages have no role in optimization
Equal-total rankings differ from equal-online rankings	Initialization cost changes the preferred tested pipeline	A free, deployable stopping or checkpoint-selection rule exists
BC is cheap and rankings do not change	The allocation tradeoff is weak at this scale	The tradeoff cannot matter at larger model or data scales
Pass^4 stays at a sampling floor	The experiment cannot resolve fine reliability differences in that regime	True reliability is zero or a method comparison is settled

The study is not powered to fit a universal model of RL readiness. The arithmetic environment verifies full traces but is artificial; horizon is coupled to branching and prompt length. The GSM8K experiment verifies answers but not all reasoning and may inherit pretraining contamination. LoRA limits the adaptation space; no full-fine-tuning comparison is funded. The core GRPO configuration has no KL penalty, replay, adaptive curriculum, or repeated clipped updates. These are explicit scope limits, not negative findings about the omitted alternatives.

Final figures will include: initial-to-final metric changes; learning curves against online and total dollars; per-horizon coverage-reliability plots; mixed-group counts and successes per dollar; the independent problem-group analysis; and the smaller GSM8K trend comparison. No unobserved result will be drawn as if measured. Analytic figures will be labeled as such.

The reproducibility package will contain the generator and acceptance rules, verifier tests, prompt templates, exact split hashes, model and package revisions, adapter configuration, seeds, loss-unit tests, training and evaluation scripts, raw attempt-level outcomes, and both cost ledgers. Unit tests must cover number multiplicity, subtraction order, negative intermediates, illegal reuse, malformed traces, alternate valid solutions, truncation, and exact numerical answer parsing. Verify loss signs, zero-group handling, adapter updates, and checkpoint restoration on deterministic toy batches.

The two students can divide ownership between environment/data/evaluation and training/infrastructure/cost logging, with mutual review of the verifier and loss implementation. Team names, outside collaboration, overlapping coursework, and applicable AI-assistance

disclosures must be supplied accurately before submission; this proposal does not invent identities or declare absent conflicts on the team's behalf.

12. Review record and status of claims

This proposal distinguishes four kinds of statements: cited research findings, analytic calculations, proposed design choices, and quantities that remain to be measured. Exact-protocol zero-shot performance, throughput, and final learning outcomes belong to the last category. All 36 branches are cost-limited experiments, not promised converged models.

Two post-draft reviews were completed. The technical review checked source-to-claim alignment, estimator formulas, horizon and zero-shot terminology, loss normalization, run counts, split design, cost arithmetic, rare-event uncertainty, and causal interpretation. It added an explicit token-level formulation, clarified inactive clipping in the one-update regime, and verified numerical illustrations independently. The editorial review clarified the execution sequence, distinguished measured quantities from assumptions, removed ambiguous terminology, and organized the protocol for readers unfamiliar with the project. The PDF was then rendered and checked for readable equations, tables, links, and pagination.

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Comprehensive ablation and control table

The table is the complete scope ledger. **Core** and **targeted** rows are funded comparisons; **diagnostic** rows reuse outputs or the evaluation allocation; **fixed** rows are controls, not sweeps; **conditional** rows are development-only decisions made before the main test protocol is frozen. Rows referring to the same branches do not add runs to the 36-branch matrix.

Experimental knob	Tested values or fixed setting	Status	Purpose	What the comparison reveals or cannot reveal
Task-specific BC exposure	0, 0.25, 2 epochs	Core A	Create different initial success profiles	Practical exposure effect; unique-example count and optimization duration are coupled
Demonstrated expression structures	Broad mixture versus sequential chains; both 2 epochs	Core B	Intervene on demonstrated solution structures	Downstream effect of the specified data restriction; not pure reliability causation
Online objective	Group-relative GRPO versus positive-only filtered supervision	Core A/B	Compare use of relative reward and successful examples	Which objective works better from each initialization under the fixed recipe
Continued offline learning	Original BC demonstrations only; no online samples	Core C	Check whether online generation is necessary at this budget	Added value of online learning relative to continued supervision
Decision horizon	2, 4, 6 operations, jointly trained	Core strata	Compare behavior as required decisions increase	Horizon-associated differences; search space and prompt length also change
Group size	G=4 versus G=8 from broad light BC	Targeted D	Test rollout allocation	Larger groups versus more distinct prompts, not a completely isolated group effect
Responses per update	128 for both objectives and G settings	Fixed	Match rollout count per update	Dollar and token differences remain visible; successful-example counts need not match
Distinct prompts per update	32 at G=4; 16 at G=8	Coupled D	Make the group-size tradeoff explicit	Prompt diversity changes with G at fixed response count
Cost convention	Equal online cost versus equal total pipeline cost	Core analysis	Charge for obtaining the initialization	Whether an apparently stronger initialization is worth its cost
Arithmetic online cap	$8 per branch; $8 total-cost checkpoint when feasible	Fixed / main	Bound spending and define estimand	Performance at the stated budget, not convergence or asymptotic superiority
Intermediate cost points	Validation near $1, $2, $4, $6 online spend	Diagnostic	Observe learning curves without exhaustive test sampling	Timing of improvement or collapse; not test-based checkpoint selection

Experimental knob	Tested values or fixed setting	Status	Purpose	What the comparison reveals or cannot reveal
Training seeds	11, 22, 33; paired conditions	Core replication	Expose optimization and rollout variability	Limited seed consistency; not high-powered significance for small effects
Domain	Arithmetic primary; GSM8K initialization-only check	Targeted E	Examine transfer of an initialization trend	Natural-language agreement or limitation; no robotics or full domain factorial
GSM8K BC / online cap	0.25 versus 2 epochs; GRPO; $6 per online branch	Targeted E	Small external check with three seeds	Initialization trend only; no GSM8K objective or coverage comparison
Primary metrics	Pass@1, pass@4, pass^4	Core measurement	Separate performance, retry coverage, and consistency	Metric-dependent tradeoffs, not independent causal axes
Auxiliary reliability order	Pass^2 from the same samples	Diagnostic	Interpret a pass^4 floor	Lower-order consistency; does not replace the declared primary metrics
Attempts per problem	n=8 primary; n=16 on a preselected 96-problem arithmetic panel if affordable	Diagnostic / conditional	Check repeated-sampling precision	Sensitivity to sampling noise; rare events can remain unresolved
Arithmetic test allocation	96 problems per horizon, 288 total	Fixed	Equal horizon weighting	Task-sampling uncertainty; no guarantee of detecting small differences
GSM8K test allocation	256 official test problems	Fixed	Affordable external evaluation	Subset performance with uncertainty, not a full-benchmark leaderboard result
Prompt demonstrations	Zero worked examples versus one format example in development; one fixed example for arithmetic main	Diagnostic/fixed	Distinguish zero-shot failure from format assistance	Main I_0 is no task-specific BC, but not zero-shot prompting
GSM8K prompting	Instructions with no worked solutions	Fixed	Define the natural-language zero-example protocol	Must be measured directly; official 49.6 is not its verified baseline
Decoding	Temperature 1, top-p 1, no top-k filter; greedy only as auxiliary accuracy audit	Fixed / diagnostic	Define the sampled policy and repeat independence	Reliability under this decoding procedure, not optimized inference settings
Completion caps	Arithmetic 256, GSM8K 512; development-only adjustment if truncation is material	Fixed / conditional	Bound cost without silently truncating valid solutions	Cap sensitivity and failures; a changed cap requires a revised common protocol

Experimental knob	Tested values or fixed setting	Status	Purpose	What the comparison reveals or cannot reveal
Reward	Binary terminal success; no format reward or partial credit	Fixed	Preserve sparse outcome supervision	Results do not test reward shaping or process-reward advantages
Verifier	Full arithmetic trace; GSM8K final numeric answer	Fixed	Use exact task-appropriate correctness	Full operation validity in arithmetic; no guaranteed GSM8K reasoning validity
Main model	Qwen2.5-0.5B-Instruct, pinned revision	Fixed	Hold architecture and pretraining constant	Within-model findings only; no funded model-scale ablation
Larger model option	1.5B discussed as future alternative, not scheduled	Not funded	Avoid an unsupported automatic fallback	No scale claim; changing model requires re-scoping the budget
LoRA	Rank 16, alpha 32, dropout 0; q/k/v/o and gate/up/down projections	Fixed	Limit trainable state consistently	Adapter-constrained learning; no rank or full-fine-tuning comparison
BC learning rate / optimizer	AdamW, 1e-4 initial rate, weight decay 0, gradient norm cap 1	Fixed / calibration	Specify a reproducible starting recipe	Not a finding about optimal supervised hyperparameters
Online learning rate	Development candidates 5e-6 and 1e-5; one frozen choice per method	Conditional	Give both methods limited equal calibration	Avoid per-initialization tuning; report discarded numerical failures
GRPO reward normalization	Group mean and population group std plus 1e-6	Fixed	Define relative advantages precisely	Findings do not generalize automatically to batch-normalized or unscaled rewards
Online loss denominator	Constant L_{max}; GRPO and filtered losses explicitly implemented	Fixed adjustment	Avoid inverse length weighting	Not the full Dr. GRPO recipe; response-length effects still need reporting
KL penalty / critic	Beta=0; no reference penalty or learned critic	Fixed	Isolate the stated reference-free objectives	No conclusion about KL-based retention or value-based methods
PPO-style clipping	Radius 0.2; one full-batch update per rollout	Fixed	Preserve explicit implementation semantics	Clipping is inactive at the initial ratio-one gradient; no trust-region claim
Replay / update reuse	No replay; one optimizer step per fresh batch	Fixed	Keep online data exposure transparent	No off-policy replay or multiple-epoch optimization study
Uniform-reward groups	Retain and log; no success-based dynamic resampling	Fixed	Observe reward sparsity without hidden collection cost	Zero relative advantages are measured rather than filtered away

Experimental knob	Tested values or fixed setting	Status	Purpose	What the comparison reveals or cannot reveal
Problem-group analysis	First 4 initial trials stratify; other 4 estimate initial accuracy; fresh final trials	Diagnostic	Separate observed discovery, consolidation, and retention	Reduces regression-to-mean artifacts; does not identify true p=0 or p=1
Additional readiness diagnostics	Held-out supervised loss, entropy, valid output rate, mixed groups, tokens and successes per dollar	Diagnostic	Compare explanatory signals with prior work	Descriptive evidence only; no fitted universal readiness predictor
Data overlap / balance	Canonical problem deduplication; disjoint BC, online, validation, test; shared manifests across seeds	Fixed	Prevent leakage and unequal test mixtures	Tests generalization within the generator; not training-dataset resampling robustness
Generator acceptance	Inputs 2-20; +,-,*; bounded target/intermediates; one development-only revision if needed	Fixed / conditional	Avoid trivial or wholly uninformative task distributions	Revision must precede test freezing; no search for desired algorithm rankings
Hardware and precision	L4 or L40S chosen by measured cost; BF16; fixed backend within comparisons	Conditional / fixed	Measure real end-to-end efficiency	Dollars depend on hardware and implementation; tokens provide a secondary view
Checkpoint selection	Fixed-budget online endpoints; optional validation-selected BC baseline charged full selection cost	Fixed	Prevent retrospective refunding of training or search	Empirical best checkpoints do not constitute a free operational switching rule
Budget protection	$500 ceiling; preserve A/B/C and primary evaluation; omit D then E if needed	Fixed contingency	Prevent infeasible scope from eroding core evidence	Missing blocks are disclosed, not interpreted as negative results